



CS105.3 - Database Management Systems

26.1 - Lab Sheet 04

Task 1

A company maintains a database to store details about its departments and employees. Each department has a unique department ID, department name, and department location. Each employee has a unique employee ID, employee name, employee town, and belongs to one department.

The database contains the following tables:

Department

- DepID
- DepName
- DepLocate

Employee

- EmpID
- EmpName
- EmpTown
- EmpDepartID

Each employee is assigned to one department, and one department can have many employees.

Draw an Entity Relationship Diagram for the above scenario. Clearly show:

1. Entities
2. Attributes
3. Primary keys
4. Foreign keys
5. Relationship between Department and Employee
6. Cardinality of the relationship

Task 2

Using Microsoft Access, create a database for the company employee system described below.

A company stores information about its departments and employees. Each department can have many employees, but each employee belongs to only one department.

You are required to create the database step by step.

Step 1: Create a new Access database

Create a new database and save it as:

EmployeeDB

Step 2: Create the Department table

Create a table named **Department** with the following fields:

Field Name	Data Type	Description
DepID	Number	Primary Key
DepName	Short Text	Department name
DepLocate	Short Text	Department location

Set **DepID** as the primary key.

Step 3: Enter data into the Department table

Enter the following records:

DepID	DepName	DepLocate
1	HR	Colombo
2	Finance	Kandy
3	IT	Galle
4	Marketing	Jaffna

Step 4: Create the Employee table

Create a table named **Employee** with the following fields:

Field Name	Data Type	Description
EmpID	Short Text	Primary Key
EmpName	Short Text	Employee name
EmpTown	Short Text	Employee town
EmpDepartID	Number	Foreign Key referencing Department.DepID

Set **EmpID** as the primary key.

Step 5: Enter data into the Employee table

Enter the following records:

EmpID	EmpName	EmpTown	EmpDepartID
E001	Amal	Colombo	1
E002	Nimal	Kandy	2
E003	Kamal	Galle	3
E004	Saman	Colombo	1
E005	Ruwan	Jaffna	4
E006	Tharushi	Matara	3

Step 6: Create the relationship

Create a relationship between the two tables:

Department.DepID → Employee.EmpDepartID

Enable **Referential Integrity**.

Step 7: View the relationship diagram

Open the **Relationships** window in Microsoft Access and add both tables.

Show the relationship:

One Department has many Employees

Step 8: Save the database

Save all tables and relationships.